Yan Zhang

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Education

Vanderbilt University Nashville, TN
Master of Computer Science, GPA: 3.84/4.00 *Aug 2024 - May 2026*
Hong Kong Baptist University Hong Kong, China
BSc (Honors) in Computer Science, GPA: 3.51/4.00 *Sep 2020 - Jun 2024*
Minor in Applied Psychology

Experience

Bozhon Precision Industry Technology (SSE: 688097) May 2025 - Aug 2025
Software Engineer Intern *Suzhou, China*

- Identified data consistency bug in Apple's manufacturing line traceability system; developed application-side services for the system as part of a backend engineering team
- Designed **RESTful API** request deduplication service based on **Spring Boot** to disambiguate device retry scans from legitimate rework passes, reducing duplicate scan rate from **8–12% to 0%**
- Enforced data integrity at the **MySQL** storage layer by implementing hard-fallback constraints, handling constraint violations gracefully with automatic retry and conflict resolution logic, ensuring the service sustained **3,000+** scan events/minute peak throughput without blocking valid rework scenarios
- Configured real-time deduplication observability layer using **Prometheus** metrics and **Grafana** dashboards with **AlertManager** threshold-based alerts, deployed full service codebase via **Docker** containers on **Kubernetes** with **CI/CD** automated rollout, shifting defect detection from weekly discovery to real-time alerts with sub-minute latency

ChangePlusPlus Oct 2024 - May 2025
Software Developer *Nashville, TN*

- Developed a privacy-preserving AI knowledge-management platform for Pennsylvania Women Work as part of a ChangePlusPlus nonprofit engineering team, enabling career coaches to convert consented session recordings into structured, searchable training resources
- Built clip-level audio consent system in **Node.js**, deployed on **AWS Lambda** with **S3**-backed storage, implementing least-privilege access and time-bound presigned URL authorization for session segment review, with consent decisions persisted to **PostgreSQL** across **1,500+** annual participants
- Designed privacy-preserving knowledge pipeline using **Whisper** and **GPT-4o t**o extract reusable insights into a vector-searchable database, reducing manual curation time from **~60** to **~10** minutes per session while limiting raw-transcript retention to **24 hours** after extraction to mitigate re-identification exposure
- Implemented **LangChain**-based **RAG** pipeline over vectorized coaching methodology records, improving top-3 retrieval relevance from **~65%** to **~88%** on coach-labeled test queries through metadata filtering, similarity-score thresholding, and prompt grounding

Projects

Cairn — Self-Evolving Agentic Harness May 2026 - Aug 2026
Python, AsyncIO, OpenAI/Anthropic SDK, MCP, Docker, JSON-RPC, Skills

- Built Python agentic harness turning a hosted **LLM** (Anthropic/OpenAI-compatible APIs) into a local agent that safely executes software-engineering tasks, handling tool-call parsing, permission checks, execution, and context compression in a closed reasoning loop
- Designed three-tier structured memory folding mechanism that reorganizes long-horizon task history into task progress, current goal and blockers, and tool usage experience, letting the agent fold context and resume from a clean state when the window fills or tool calls repeatedly fail
- Designed self-evolving skill pipeline that extracts reusable rules from user feedback and applies **add/merge/discard** decisions to persist as **SKILL.md**, keeping self-modification auditable through provenance, version snapshots, and usage stats

Healthcare Care Plan Automation System Jan 2026 - Mar 2026
Python, Django, React, PostgreSQL, Docker, AWS (Lambda · SQS · RDS), Terraform, Redis

- Designed **event-driven architecture** replacing synchronous LLM calls with async task queue (**Celery + Redis → AWS SQS + Lambda**) with **Terraform**-managed infrastructure, preventing server timeouts and supporting concurrent order submission
- Built **RESTful API** with **Django REST** Framework implementing multi-layer duplicate detection (patient, provider, order) with configurable ERROR/WARNING logic and unified exception handling
- Implemented Adapter Pattern to normalize JSON/XML clinical orders into a unified `InternalOrder` schema covering patient, provider, medication, and diagnosis fields, with per-source adapters handling parsing and field mapping without changing core business logic

Skills

Languages: Python, Java, C/C++, JavaScript, SQL, TypeScript, Go, HTML/CSS
Frameworks: SpringBoot, Django, ASP.NET, React, Node.js, Next.js, MUI, Tailwind CSS, JQuery
Tools: Git, Docker, Redux, AWS (S3, Lambda, EC2), Redis, Message Queue, Nginx, CI/CD Pipeline, OAuth2, JWT
Database management systems: MySQL, PostgreSQL, MongoDB